not proportional to Γ^*, the asymptotic variances $\sigma^2_\psi(\Gamma)$ and $\sigma^2_{\phi^d}(\Gamma)$ are strictly larger than for Γ^*.

We summarize the intuition behind proposition 3. Each entity i affects the price proportionally to its size S_i; see (1). Hence, the economically appropriate weights are S. However, we need to satisfy $\Gamma'\lambda = 0$ to remove the influence from aggregate shocks on the GIV. Proposition 3 shows that the optimal weights vector is Γ^*; it is the vector closest to S, while being orthogonal to the factor loadings λ.

C. *Controlling for Observed and Latent Factors*

So far, we used z_t only to estimate the elasticities without accounting for observed and latent factors. We now show how to account for those in the procedure.

We proceed in two steps. First, in section II.C.1, we show that we can rewrite the demand equation in (2) in a more convenient form:

$$y_{it} = \phi^d p_t + C^y_{it} m^y + \eta_{1t} + \check{\lambda}_i \check{\eta}_t + \check{u}_{it}, \tag{18}$$

where we replace $\lambda\eta_t = \eta_{1t} + \lambda_i\eta_t$ by $\eta_{1t} + \check{\lambda}_i\check{\eta}_t$ and replace u_{it} by $\check{u}_{it}$, where $\check{\eta}_t$ and $\check{u}_{it}$ are defined below. This reformulation has two advantages. First, when m^y and $\check{\lambda}_i$ are known, $\check{\eta}_t$ and $\check{u}_{it}$ can be recovered without error, unlike the η_t and u_{it} of the original factor model (2). Second, it holds that $\check{\eta}_t \perp \check{u}_t$. These two convenient features are then used in proposition 4, which is the main result of this section.

1. Introducing Recoverable Factors

Before turning to this section's main result, we introduce the concept of "recoverable factors" that will simplify the idea and proofs that follow. We start from the factor model part of our system (calling it Y_t rather than y_t):

$$Y_t = \lambda\eta_t + u_t, \qquad \eta_t \perp u_t, \tag{19}$$

and we maintain assumption 1 that we know λ. In this case, η_t cannot be recovered without error when the number of entities N is fixed, as it is in our case (one would need $N \to \infty$; see, e.g., Bai 2003). The main insight is that the estimates and residuals that we can recover without error, $\check{\eta}_t$ and $\check{u}_t$, are uncorrelated, just like the true values, η_t and u_t. These estimates can be computed even though N is fixed and have useful properties that lemma 1 spells out.

LEMMA 1 (Recoverable factors and their properties). Let assumption 3 hold. Given a factor model (19), define the $r \times N$ and $N \times N$ matrices:

$$R^\lambda := (\lambda'\lambda)^{-1}\lambda', \qquad Q^\lambda := I - \lambda R^\lambda \tag{20}$$

and

$$\check{\eta}_t := \eta_t + R^{\lambda} u_t, \qquad \check{u}_t := Q^{\lambda} u_t. \tag{21}$$

Then we have

$$Y_t = \lambda \check{\eta}_t + \check{u}_t, \qquad \check{\eta}_t, \eta_t \perp \check{u}_t. \tag{22}$$

In addition, $\check{\eta}_t$ and $\check{u}_t$ can be exactly recovered from Y_t and λ, via

$$\check{\eta}_t = R^{\lambda} Y_t, \qquad \check{u}_t = Q^{\lambda} Y_t. \tag{23}$$

In the remainder of this section, we work with the recoverable shocks $(\check{\eta}_t, \check{u}_t)$ rather than the nonrecoverable shocks (η_t, u_t). Just as η_t and u_t are uncorrelated, $\check{\eta}_t$ and $\check{u}_t$ are uncorrelated as well—compare (19) and (22). However, proofs and concepts are easier working with $(\check{\eta}_t, \check{u}_t)$ rather than with (η_t, u_t), as $(\check{\eta}_t, \check{u}_t)$ are directly recovered from Y_t (see [23]).[8] By a slight abuse of language, we refer to $\check{\eta}_t$ as common factors and $\check{u}_t$ as a vector of idiosyncratic shocks.[9]

Computing $\check{\eta}_t$ and $\check{u}_t$ in lemma 1 corresponds to running cross-sectional regressions of Y_{it} on λ_i, for every date t:

$$Y_{it} = \lambda_i \check{\eta}_t + \check{u}_{it}, \tag{24}$$

which yields an exact recovery of $\check{\eta}_t$ and $\check{u}_{it}$.

2. GIV, Controlling for Observable and Recovered Controls

Before stating the procedure with controls, we need a bit more notation. We decompose $\eta_t = (\eta_{1t}, \eta_{2t})$, where η_t, η_{1t}, and η_{2t} have dimensions r, 1, and $(r-1)$, respectively. We apply the idea of lemma 1 to $\check{\lambda}$ (recall that we decomposed $\lambda_i = (1, \check{\lambda}_i)$ with $\iota'\check{\lambda} = 0_{1\times(r-1)}$), and hence we define

$$\check{\eta}_t := R^{\check{\lambda}}(\eta_t + \lambda u_t) = \eta_{2t} + R^{\check{\lambda}} u_t, \qquad \check{u}_t := Q^{\lambda} u_t. \tag{25}$$

Then, we can recover $\check{\eta}_t$ and $\check{u}_t$, and we still have $\check{\eta}_t \perp \check{u}_t$.[10] We decompose $\eta_{1t} = b^1 \check{\eta}_t + \eta_{1t}^{\perp}$ and $\varepsilon_t = b^{\varepsilon}\check{\eta}_t + \varepsilon_t^{\perp}$, where $\eta_{1t}^{\perp}$ and $\varepsilon_t^{\perp}$ are uncorrelated with $\check{\eta}_t$; they are the residuals after projecting on $\check{\eta}_t$. We call $\check{y}_{it} = y_{it} - y_{\bar{E}t}$ the cross-sectionally demeaned value with quasi-equal weights $\bar{E}$ satisfying (10). Defining $b^y := \check{\lambda}_{\bar{E}} + b^1$ and $\varepsilon_t^y := \eta_{1t}^{\perp} + u_{\bar{E}t}$, we write $y_{\bar{E}t}$ as

$$y_{\bar{E}t} = \phi^d p_t + b^y \check{\eta}_t + C_{\bar{E}t}^y m^y + \varepsilon_t^y. \tag{26}$$

[8] Even when λ is not known, $(\check{\eta}_t, \check{u}_t)$ are still very useful. See, e.g., proposition 7.

[9] Even if the u_{it}'s are uncorrelated across i's, the $\check{u}_{it}$'s will have some slight correlation across i's since $V^{\check{u}} = Q\sigma_u^2$. This sometimes requires a bit of care, which we take.

[10] Running the regression $y_{it} - C_{it}^y m^y = a_t + \check{\lambda}_i \check{\eta}_t + \check{u}_{it}$ on the known $\check{\lambda}_i$ yields $\check{\eta}_t$ and $\check{u}_{it}$; see (28). This holds because $\check{\eta}_t = R^{\check{\lambda}}(y_t - C_t^y m^y)$, $\check{u}_t = Q^{\check{\lambda}}(y_t - C_t^y m^y)$.

PROPOSITION 4 (GIV estimation with controls). Let assumptions 1 and 3 hold. Define $\check{y}_{it} := y_{it} - y_{Et}$ and similarly $\check{C}^y_{it} := C^y_{it} - C^y_{Et}$. Given a candidate value m^y, we construct $\check{\eta}_t(m^y) := R^{\check{\lambda}}(y_t - C^y_t m^y)$ and the GIV

$$z_t(m^y) := \Gamma'(y_t - C^y_t m^y), \tag{27}$$

with Γ satisfying (4). Define θ to be the collection of parameters $(\psi, \phi^d, m^p, b^p, m^y, b^y)$ and θ_0 the true value of θ. Define θ^e_T to be the estimator of θ_0 that solves the following sample moments:

$$\sum_t \left(-\check{y}_t + \check{\lambda}\check{\eta}_t(m^y) + \check{C}^y_t m^y\right)' \check{C}^y_t = 0, \tag{28}$$

$$\sum_t \left(-p_t + \psi y_{St} + b^p \check{\eta}_t(m^y) + C^p_t m^p\right)\left(z_t(m^y), \check{\eta}_t(m^y)', C^{p\prime}_t\right) = 0. \tag{29}$$

To estimate ϕ^d (assuming that $\psi \neq 0$), we use the additional moment condition

$$\sum_t \left(-y_{\tilde{E}t} + \phi^d p_t + b^y \check{\eta}_t(m^y) + C^y_{\tilde{E}t} m^y\right)\left(z_t(m^y), \check{\eta}_t(m^y)', C^{y\prime}_{\tilde{E}t}\right) = 0. \tag{30}$$

Under the additional regularity assumption 5 from appendix A, θ^e_T is consistent and $\sqrt{T}$-asymptotically normal. Assuming that $\varepsilon^\perp_t$ and ε^y_t are homoskedastic conditionally on u_t, the variances of the estimators ψ^e_T and $\phi^{d,e}_T$ are as in proposition 2, with the new values $\sigma^2_{\varepsilon^\psi} = \mathrm{var}(\varepsilon^\perp_t)$ and $\sigma^2_{\varepsilon^\phi} = \mathrm{var}(\varepsilon^y_t)$. In addition, the optimal Γ remains as in proposition 3.

The upshot of proposition 4 is that we can estimate the parameters as in our basic proposition 2, except that the controls soak up more variance.[11] In addition, under assumption 1, the standard errors of ψ^e_T and $\phi^{d,e}_T$ are the asymptotic standard errors that a "naive" IV estimator would report that ignores that m^y and $\check{\eta}_t$ have been estimated.

In summary, when there are no controls and under the assumptions of proposition 4, the optimal GIV is

$$z_t := \Gamma^{*\prime} y_t = S' Q y_t = S' \check{u}_t, \tag{31}$$

with $\check{u}_t := Q u_t$. In other terms, $\check{u}_{it}$ is the residual from a cross-sectional regression of $\check{y}_{it} := y_{it} - y_{Et}$, on the demeaned factor loadings $\check{\lambda}_i$:

$$\check{y}_{it} := \check{\lambda}_i \check{\eta}_t + \check{u}_{it}. \tag{32}$$

This regression also recovers the aggregate shocks $\check{\eta}_t$; see lemma 1. When there are controls, we simply replace y_t by $y_t - C^y_t m^y$.

[11] The assumption $\psi \neq 0$ is needed only to estimate ϕ^d via (28) and (30). If $\psi = 0$, we can still estimate ψ by the moments (28) and (29).

D. *Using the GIV via OLS*

So far, we have used the GIV z_t in an IV form. We now show how to use it in estimating μ and M using OLS, where we defined μ and M in (13). First, we solve for y_{St}, which leads to

$$p_t = \mu u_{St} + b^p c_t + \varepsilon_t^p, \tag{33}$$

$$y_{St} = M u_{St} + b^y c_t + \varepsilon_t^y, \tag{34}$$

where $c_t = (C_t^p, C_{St}^y, \check{\eta}_t)$ is a vector of controls and $(\varepsilon_t^p, \varepsilon_t^y)$ are uncorrelated with u_{St}. If y_{it} represents log demand and p_t represents the log price, the interpretation is that a 1% demand shock leads to a μ% price increase and an M% supply increase. The intuition is that the size-weighted idiosyncratic shock u_{St} affects p_t with a strength μ and y_{St} affects p_t with a strength M.

Next, we introduce Γ satisfying (4) and such that

$$\frac{\mathbb{E}[u_{St} u_{\Gamma t}]}{\mathbb{E}[u_{\Gamma t}^2]} = 1. \tag{35}$$

Defining $\tilde{E} := S - \Gamma$, condition (35) is equivalent to having (10).[12] Then $u_{St} = u_{\Gamma t} + u_{\tilde{E}t}$ and

$$p_t = \mu z_t + b^p c_t + e_t^p, \tag{36}$$

$$y_{St} = M z_t + b^y c_t + e_t^y, \tag{37}$$

with $e_t^p := \varepsilon_t^p + \mu u_{\tilde{E}t}$ and $e_t^y := \varepsilon_t^y + M u_{\tilde{E}t}$, which are orthogonal to z_t and c_t (recall [10]). It follows that we can estimate μ and M by an OLS regression of p_t and y_{St} on z_t.[13] The next proposition records that result and derives the precision of the estimators.

PROPOSITION 5 (GIV estimator for OLS). Let assumptions 1 and 3 hold, and assume (35). Consider the procedure that (i) estimates m^y by the regression (28), where $\check{\eta}_t(m^y) := R^{\check{\lambda}}(y_t - C_t^y m^y)$ and $R^{\check{\lambda}} = (\check{\lambda}'\check{\lambda})^{-1}\check{\lambda}'$; (ii) defines the GIV z_t by (27); and (iii) estimates μ and M by the OLS regression coefficients of (36) and (37), respectively, where $c_t = (C_t^p, C_{St}^y, \check{\eta}_t(m^y))$ is a vector of controls. Then, under the additional regularity assumption 5 from appendix A, the OLS estimators μ_T^e and M_T^e are consistent. In addition, $\sqrt{T}(\mu_T^e - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma_\mu^2)$ and $\sqrt{T}(M_T^e - M) \xrightarrow{d} \mathcal{N}(0, \sigma_M^2)$. Assuming that e_t^p and e_t^y are homoskedastic conditionally on

[12] So, e.g., we can use $\Gamma = S - \tilde{E}$ with $\widetilde{E}_i = 1/N$ in the homoskedastic case and (54) in the heteroskedastic case.

[13] Alternatively, in the regressions (36) and (37) one can replace z_t by $Z_t := y_{St} - y_{\tilde{E}t}$. The result is then the same, including the standard error of the estimators. This is simply because with $z_t = Z_t - \check{\lambda}_\Gamma \check{\eta}_t$, we control for $\check{\eta}_t$.